

27.5 Stein's paradox and James–Stein estimation

The shrinkage estimators so far beat least squares only for *some* parameter values, which is unavoidable for linear estimators (Exercise (B1)). Stein (1956) discovered that a *nonlinear* shrinkage can do better everywhere: when three or more means are estimated together under total squared error, the estimator that is at once least squares, maximum likelihood and best unbiased is inadmissible. James and Stein (1961) gave an explicit estimator that dominates it. This section proves their theorem and explains it as an empirical Bayes phenomenon.

27.5.1 The normal means problem

Let $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ with σ^2 known, and measure an estimator $\hat{\boldsymbol{\theta}}(\mathbf{Z})$ by its **risk** $R(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}) = E\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2$, its total mean squared error. The estimator $\mathbf{Z}$ has risk $p\sigma^2$ for every $\boldsymbol{\theta}$. An estimator **dominates** another if its risk is never larger and is smaller for some $\boldsymbol{\theta}$; an estimator dominated by none is **admissible**.

This is regression in canonical form. If $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, then $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ (Theorem 7.5.1(a)). For a general full-rank design, $\mathbf{Z} = (\mathbf{X}^\top \mathbf{X})^{1/2} \hat{\boldsymbol{\beta}}$ is $N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\theta} = (\mathbf{X}^\top \mathbf{X})^{1/2} \boldsymbol{\beta}$, and for any estimator $\tilde{\boldsymbol{\beta}}$ with $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{1/2} \tilde{\boldsymbol{\beta}}$,

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2 = (\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{X}^\top \mathbf{X} (\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \|\mathbf{X}\tilde{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta}\|^2,$$

the prediction loss at the design points, with the square root of Theorem 1.7.6 as the change of coordinates.

27.5.2 Stein's lemma

The proof rests on an integration-by-parts identity for the normal distribution.

Lemma 27.5.1. Stein's lemma

- (a) Let $Z \sim N(\mu, \sigma^2)$, and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be absolutely continuous with $E|g'(Z)| < \infty$. Then $E|(Z - \mu)g(Z)| < \infty$ and

$$E[(Z - \mu)g(Z)] = \sigma^2 E g'(Z).$$

- (b) Let $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$, and let $\mathbf{g} = (g_1, \dots, g_p) : \mathbb{R}^p \rightarrow \mathbb{R}^p$ be such that, for each i and almost every value of the other coordinates, g_i is absolutely continuous in z_i , with $E|\partial g_i(\mathbf{Z})/\partial z_i| < \infty$ and $E|(Z_i - \theta_i)g_i(\mathbf{Z})| < \infty$. Then

$$E[(\mathbf{Z} - \boldsymbol{\theta})^\top \mathbf{g}(\mathbf{Z})] = \sigma^2 E \sum_{i=1}^p \frac{\partial g_i}{\partial z_i}(\mathbf{Z}). \quad (27.5.1)$$

Proof.

- (a) Write $Z = \mu + \sigma W$ with $W \sim N(0, 1)$ and $h(w) = g(\mu + \sigma w)$, so that $h'(w) = \sigma g'(\mu + \sigma w)$. The claim becomes $E[Wh(W)] = E h'(W)$, so we may take $\mu = 0$, $\sigma = 1$,

$g = h$. Let φ be the standard normal density. Since $\varphi'(t) = -t\varphi(t)$,

$$\begin{aligned}\varphi(w) &= \int_w^\infty t\varphi(t) dt \quad (w \geq 0), \\ \varphi(w) &= -\int_{-\infty}^w t\varphi(t) dt \quad (w \leq 0).\end{aligned}$$

Hence

$$\begin{aligned}E h'(W) &= \int_0^\infty h'(w) \int_w^\infty t\varphi(t) dt dw \\ &\quad - \int_{-\infty}^0 h'(w) \int_{-\infty}^w t\varphi(t) dt dw.\end{aligned}$$

The double integrals converge absolutely: replacing h' by $|h'|$ and t by $|t|$ gives $\int |h'(w)|\varphi(w) dw = E|h'(W)| < \infty$. By Fubini's theorem we may reverse the order of integration. The first term becomes $\int_0^\infty t\varphi(t) \int_0^t h'(w) dw dt = \int_0^\infty t\varphi(t)\{h(t) - h(0)\} dt$, and the second $-\int_{-\infty}^0 t\varphi(t) \int_t^0 h'(w) dw dt = \int_{-\infty}^0 t\varphi(t)\{h(t) - h(0)\} dt$, using absolute continuity of h . Their sum is $E[W\{h(W) - h(0)\}] = E[Wh(W)]$, since $EW = 0$. The same bound shows $E|W\{h(W) - h(0)\}| \leq E|h'(W)|$, so $E|Wh(W)| < \infty$.

- (b) The coordinates of $\mathbf{Z}$ are independent. Fix i and condition on the other coordinates $\mathbf{Z}_{-i}$. By Fubini, $E[|\partial g_i / \partial z_i| \mid \mathbf{Z}_{-i}] < \infty$ for almost every value of $\mathbf{Z}_{-i}$, and for those values part (a), applied to $z_i \mapsto g_i(\mathbf{Z})$ with $Z_i \sim N(\theta_i, \sigma^2)$, gives $E[(Z_i - \theta_i)g_i(\mathbf{Z}) \mid \mathbf{Z}_{-i}] = \sigma^2 E[\partial g_i(\mathbf{Z}) / \partial z_i \mid \mathbf{Z}_{-i}]$. Take expectations and sum over i .

■

The lemma turns an expectation involving the unknown $\boldsymbol{\theta}$ into the expectation of a function of $\mathbf{Z}$ alone. Stein (1981) built a theory of risk estimation on it ([Exercise \(C2\)](#)), and it computes covariance degrees of freedom ([Exercise \(C1\)](#)) for nonlinear fits such as the lasso ([Exercise \(C1\)](#)).

27.5.3 The James–Stein theorem

Theorem 27.5.2. James–Stein

Let $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ with $p \geq 3$ and σ^2 known, and for a constant c let $\hat{\boldsymbol{\theta}}_c = (1 - c\sigma^2/\|\mathbf{Z}\|^2)\mathbf{Z}$. Then $E\|\mathbf{Z}\|^{-2} < \infty$ and:

- (a) $E\|\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}\|^2 = p\sigma^2 - c\{2(p-2) - c\}\sigma^4 E\|\mathbf{Z}\|^{-2}$. This is below $p\sigma^2$ for every $\boldsymbol{\theta}$ iff $0 < c < 2(p-2)$, and smallest at $c = p-2$. The **James–Stein estimator** is

$$\hat{\boldsymbol{\theta}}^{\text{JS}} = \left(1 - \frac{(p-2)\sigma^2}{\|\mathbf{Z}\|^2}\right)\mathbf{Z},$$

with risk

$$E\|\hat{\boldsymbol{\theta}}^{\text{JS}} - \boldsymbol{\theta}\|^2 = p\sigma^2 - (p-2)^2\sigma^4 E\|\mathbf{Z}\|^{-2} < p\sigma^2. \quad (27.5.2)$$

- (b) With $\gamma = \|\boldsymbol{\theta}\|^2/\sigma^2$ and $K \sim \text{Poisson}(\gamma/2)$, $\sigma^2 E\|\mathbf{Z}\|^{-2} = E\{1/(p-2+2K)\}$. The risk of $\hat{\boldsymbol{\theta}}^{\text{JS}}$ depends on $\boldsymbol{\theta}$ only through γ , equals $2\sigma^2$ at $\boldsymbol{\theta} = \mathbf{0}$, increases strictly in γ , and tends to $p\sigma^2$ as $\gamma \rightarrow \infty$.

Proof. Finiteness. The density of $\mathbf{Z}$ is at most $(2\pi\sigma^2)^{-p/2}$, so

$$\begin{aligned} \mathbb{E}\|\mathbf{Z}\|^{-2} &\leq (2\pi\sigma^2)^{-p/2} \int_{\|\mathbf{z}\|\leq 1} \|\mathbf{z}\|^{-2} d\mathbf{z} + 1 \\ &= (2\pi\sigma^2)^{-p/2} A_p \int_0^1 r^{p-3} dr + 1 < \infty, \end{aligned}$$

where A_p is the surface area of the unit sphere in $\mathbb{R}^p$ and the integral is finite because $p \geq 3$. (a) Apply Eq. 27.5.1 to $\mathbf{g}(\mathbf{z}) = \mathbf{z}/\|\mathbf{z}\|^2$. For $p \geq 3$ the other coordinates $\mathbf{Z}_{-i}$ are nonzero with probability one, and then g_i is a smooth function of z_i with

$$\frac{\partial g_i}{\partial z_i} = \frac{1}{\|\mathbf{z}\|^2} - \frac{2z_i^2}{\|\mathbf{z}\|^4}, \quad \left| \frac{\partial g_i}{\partial z_i} \right| \leq \frac{3}{\|\mathbf{z}\|^2}, \quad \sum_i \frac{\partial g_i}{\partial z_i} = \frac{p-2}{\|\mathbf{z}\|^2}.$$

The bound makes $\mathbb{E}|\partial g_i/\partial z_i|$ finite, and $|(Z_i - \theta_i)g_i(\mathbf{Z})| \leq |Z_i - \theta_i|/\|\mathbf{Z}\|$ has finite expectation by the Cauchy–Schwarz inequality. So $\mathbb{E}[(\mathbf{Z} - \boldsymbol{\theta})^\top \mathbf{Z}/\|\mathbf{Z}\|^2] = (p-2)\sigma^2 \mathbb{E}\|\mathbf{Z}\|^{-2}$. Now expand

$$\|\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}\|^2 = \|\mathbf{Z} - \boldsymbol{\theta}\|^2 - 2c\sigma^2 \frac{(\mathbf{Z} - \boldsymbol{\theta})^\top \mathbf{Z}}{\|\mathbf{Z}\|^2} + \frac{c^2\sigma^4}{\|\mathbf{Z}\|^2}$$

and take expectations: $p\sigma^2 - 2c(p-2)\sigma^4 \mathbb{E}\|\mathbf{Z}\|^{-2} + c^2\sigma^4 \mathbb{E}\|\mathbf{Z}\|^{-2}$. The factor $c\{2(p-2) - c\}$ is positive iff $0 < c < 2(p-2)$ and largest at $c = p-2$, where it equals $(p-2)^2$. (b) $\mathbf{Z}/\sigma \sim N_p(\boldsymbol{\theta}/\sigma, \mathbf{I})$, so $U = \|\mathbf{Z}\|^2/\sigma^2 \sim \chi^2(p, \gamma)$ (Definition 4.1.1), and by Theorem 4.1.2(d), given $K = k$, $U \sim \chi^2(p+2k)$. For $m \geq 3$, $\mathbb{E}\{1/\chi^2(m)\} = 1/(m-2)$ (Exercise (A1)), which gives the formula. At $\gamma = 0$, $K = 0$ and the risk is $p\sigma^2 - (p-2)\sigma^2 = 2\sigma^2$. Since $\mathbb{E}(1/U) = \int_0^\infty \Pr(U < 1/s) ds$ and each $\Pr(U < 1/s)$ decreases strictly in γ (Proposition 4.1.3), $\mathbb{E}(1/U)$ decreases strictly and the risk increases. As $\gamma \rightarrow \infty$, $K \rightarrow \infty$ in probability and $1/(p-2+2K) \leq 1/(p-2)$, so $\mathbb{E}\{1/(p-2+2K)\} \rightarrow 0$ by dominated convergence. ■

The shrinkage factor is close to one when $\|\mathbf{Z}\|^2$ is large compared with its null expectation $p\sigma^2$, and small when the data are consistent with $\boldsymbol{\theta} \approx \mathbf{0}$. Nothing is special about the origin: shrinking towards any point fixed in advance also dominates $\mathbf{Z}$, with the largest gains near that point (Exercise (B1)).

The paradox is that the coordinates need not be related. If θ_1 is a physical constant, θ_2 a crop yield and θ_3 a batting average, the James–Stein estimate of the crop yield depends on the other two measurements. The resolution is that the theorem concerns the *total* risk; a single coordinate's risk can exceed σ^2 (Exercise (B2)).

Example. The risk function for ten means

For $p = 10$ and $\sigma = 1$, the exact risk from Theorem 27.5.2(b) is 2 at $\boldsymbol{\theta} = \mathbf{0}$, 4.784 at $\|\boldsymbol{\theta}\|^2 = 5$, 7.587 at 20 and 9.257 at 80, always below the constant risk 10 of $\mathbf{Z}$ (Figure 27.5.1). The fixed shrinkage $\mathbf{Z}/(1+0.5)$, which is ridge regression with $\lambda = 0.5$ in an orthonormal design, has risk $(p + 0.25\|\boldsymbol{\theta}\|^2)/2.25$ (Exercise (A2)). It beats James–Stein only on a middle range, $5.9 < \|\boldsymbol{\theta}\|^2 < 36.6$. At the origin it is worse ($10/2.25 = 4.44$ against 2), it exceeds p once $\|\boldsymbol{\theta}\|^2 > 5p = 50$, and it grows without bound. The dotted curve is the risk $p\|\boldsymbol{\theta}\|^2/(p + \|\boldsymbol{\theta}\|^2)$ of the best linear shrinkage $f\mathbf{Z}$ for each $\boldsymbol{\theta}$, the oracle of Proposition 27.1.2(c) with a common factor; James–Stein stays within 2 of it without knowing $\boldsymbol{\theta}$.

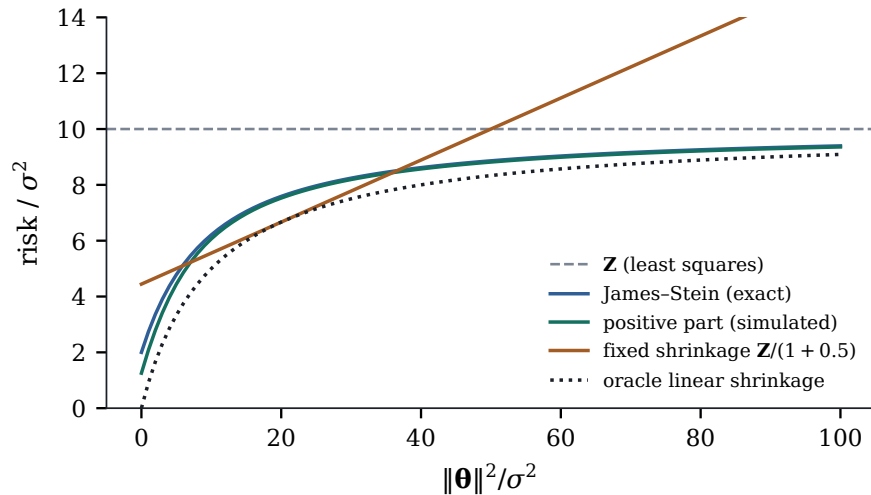


Figure 27.5.1: Risk of estimators of ten normal means with $\sigma = 1$, against $\|\boldsymbol{\theta}\|^2$: the unbiased estimator $\mathbf{Z}$, James–Stein (exact, from the Poisson mixture), its positive part (simulated, 4000 draws per point), a fixed proportional shrinkage, and the best linear shrinkage for each $\boldsymbol{\theta}$.

```
import numpy as np
from scipy import stats
p = 10

def js_risk(gamma, p=p):
    """Exact risk of  $(1 - (p-2)/\|\mathbf{Z}\|^2) \mathbf{Z}$  when  $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \mathbf{I})$  and  $\gamma = \|\boldsymbol{\theta}\|^2$ ."""
    k = np.arange(0, 400)
    w = stats.poisson.pmf(k, gamma / 2)          #  $\|\mathbf{Z}\|^2 \sim \chi^2(p, \gamma)$ :
    Poisson(gamma/2) mixture
    return p - (p - 2) ** 2 * np.sum(w / (p - 2 + 2 * k))

def js(z, positive=False):
    factor = 1 - (len(z) - 2) / np.sum(z ** 2)
    return (max(factor, 0.0) if positive else factor) * z

rng = np.random.default_rng(27)
for gamma in [0.0, 5.0, 20.0, 80.0]:
    theta = np.sqrt(gamma / p) * np.ones(p)
    Zs = theta + rng.normal(size=(2_000, p))
    loss_js = np.mean([np.sum((js(z) - theta) ** 2) for z in Zs])
    loss_pp = np.mean([np.sum((js(z, True) - theta) ** 2) for z in Zs])
    print(f"||theta||^2 = {gamma:5.1f}: exact JS risk {js_risk(gamma):.3f}, "
          f"simulated {loss_js:.3f}, positive part {loss_pp:.3f} (MLE: {p})")
```

[Run this code online](#)

27.5.4 The positive part

When $\|\mathbf{Z}\|^2 < (p-2)\sigma^2$ the James–Stein factor is negative, and the estimator reverses the sign of every coordinate, which is absurd. Truncating the factor at zero helps.

Proposition 27.5.3. Positive-part James–Stein

In the setting of [Theorem 27.5.2](#), the estimator

$$\hat{\boldsymbol{\theta}}^{\text{JS}+} = \left(1 - \frac{(p-2)\sigma^2}{\|\mathbf{Z}\|^2}\right)_+ \mathbf{Z}$$

has strictly smaller risk than $\hat{\boldsymbol{\theta}}^{\text{JS}}$ for every $\boldsymbol{\theta}$.

Proof. Write $\hat{\boldsymbol{\theta}}^{\text{JS}} = h\mathbf{Z}$ with $h = 1 - (p-2)\sigma^2/\|\mathbf{Z}\|^2$, and let $\psi = \max(-h, 0) \geq 0$, a function of $\|\mathbf{Z}\|$. Where $h \geq 0$ the estimators agree. Where $h < 0$, the positive part is $\mathbf{0}$, and

$$\|h\mathbf{Z} - \boldsymbol{\theta}\|^2 - \|\boldsymbol{\theta}\|^2 = h^2\|\mathbf{Z}\|^2 - 2h\boldsymbol{\theta}^\top \mathbf{Z} = \psi^2\|\mathbf{Z}\|^2 + 2\psi\boldsymbol{\theta}^\top \mathbf{Z}.$$

So the difference of risks is $E[\psi^2\|\mathbf{Z}\|^2] + 2E[\psi\boldsymbol{\theta}^\top \mathbf{Z}]$. The first term is positive, because $\Pr(h < 0) = \Pr\{\|\mathbf{Z}\|^2 < (p-2)\sigma^2\} > 0$. For the second, choose an orthogonal $\mathbf{Q}$ with $\mathbf{Q}\boldsymbol{\theta} = m\mathbf{e}_1$, $m = \|\boldsymbol{\theta}\|$. Then $\mathbf{w} = \mathbf{Q}\mathbf{Z} \sim N_p(m\mathbf{e}_1, \sigma^2\mathbf{I})$, $\|\mathbf{w}\| = \|\mathbf{Z}\|$ and $\boldsymbol{\theta}^\top \mathbf{Z} = mw_1$. Given $R = \sum_{i \geq 2} w_i^2$, which is independent of $w_1 \sim N(m, \sigma^2)$,

$$\begin{aligned} E[\psi w_1 \mid R] &= \int_0^\infty \psi(\sqrt{t^2 + R}) t \{\varphi_\sigma(t - m) - \varphi_\sigma(t + m)\} dt \\ &\geq 0, \end{aligned}$$

where φ_σ is the $N(0, \sigma^2)$ density: the integrand pairs t with $-t$, and $(t - m)^2 \leq (t + m)^2$ for $t, m \geq 0$. (The expectations are finite since $\psi \leq (p-2)\sigma^2/\|\mathbf{Z}\|^2$.) Hence $E[\psi\boldsymbol{\theta}^\top \mathbf{Z}] \geq 0$ and the risk difference is positive. ■

The positive-part estimator is itself inadmissible, but improvements on it are tiny (Lehmann and Casella 1998, chapter 5). For $p = 10$ its risk at the origin is $E[\{1 - (p-2)/U\}_+^2 U] = 1.258$ with $U \sim \chi^2(10)$, against 2 for James–Stein, and the two curves in [Figure 27.5.1](#) merge as negative factors become rare.

27.5.5 The empirical Bayes explanation

Efron and Morris (1973) explained the paradox by drawing the means themselves from a normal distribution.

Proposition 27.5.4. James–Stein as empirical Bayes

Let $\boldsymbol{\theta} \sim N_p(\mathbf{0}, \omega^2\mathbf{I})$ and $\mathbf{Z} \mid \boldsymbol{\theta} \sim N_p(\boldsymbol{\theta}, \sigma^2\mathbf{I})$, $p \geq 3$, and put $B = \sigma^2/(\sigma^2 + \omega^2)$.

- (a) The posterior mean is $E(\boldsymbol{\theta} \mid \mathbf{Z}) = (1 - B)\mathbf{Z}$, and its Bayes risk is $E\|(1 - B)\mathbf{Z} - \boldsymbol{\theta}\|^2 = p\sigma^2(1 - B)$.
- (b) Marginally $\mathbf{Z} \sim N_p(\mathbf{0}, (\sigma^2 + \omega^2)\mathbf{I})$, and $(p-2)\sigma^2/\|\mathbf{Z}\|^2$ is an unbiased estimator of B .
- (c) The Bayes risk of $\hat{\boldsymbol{\theta}}^{\text{JS}}$ is $p\sigma^2(1 - B) + 2\sigma^2 B$.

Proof.

- (a) $(\boldsymbol{\theta}, \mathbf{Z})$ is jointly normal with $\text{Cov}(\boldsymbol{\theta}, \mathbf{Z}) = \omega^2 \mathbf{I}$ and $\text{Cov}(\mathbf{Z}) = (\sigma^2 + \omega^2) \mathbf{I}$. By Theorem 3.5.2, $\boldsymbol{\theta} \mid \mathbf{Z} \sim N_p(\omega^2(\sigma^2 + \omega^2)^{-1} \mathbf{Z}, \sigma^2 \omega^2(\sigma^2 + \omega^2)^{-1} \mathbf{I})$; the mean is $(1 - B) \mathbf{Z}$, and the Bayes risk is the expected trace of the posterior covariance, $p \sigma^2 \omega^2 / (\sigma^2 + \omega^2) = p \sigma^2 (1 - B)$.
- (b) $\mathbf{Z} = \boldsymbol{\theta} + (\mathbf{Z} - \boldsymbol{\theta})$ is a sum of independent normal vectors. So $\|\mathbf{Z}\|^2 / (\sigma^2 + \omega^2) \sim \chi^2(p)$, and $E\{(p - 2) \sigma^2 / \|\mathbf{Z}\|^2\} = (p - 2) \sigma^2 / \{(\sigma^2 + \omega^2)(p - 2)\} = B$ by Exercise (A1).
- (c) Average Eq. 27.5.2 over the prior: the Bayes risk is $p \sigma^2 - (p - 2)^2 \sigma^4 E\|\mathbf{Z}\|^{-2}$ with the marginal expectation $E\|\mathbf{Z}\|^{-2} = 1 / \{(\sigma^2 + \omega^2)(p - 2)\}$, which gives $p \sigma^2 - (p - 2) \sigma^2 B = p \sigma^2 (1 - B) + 2 \sigma^2 B$. ■

James–Stein plugs an unbiased estimate of B , based on the spread of the p observations, into the Bayes rule, and part (c) shows that not knowing ω^2 costs at most $2\sigma^2$ in total. This is the **empirical Bayes** idea: the ensemble of parameters supplies its own prior. Theorem 27.5.2 says more, that the gain is guaranteed even if no such prior exists.

In an orthonormal design ridge regression is the Bayes rule $\mathbf{Z}/(1 + \lambda)$ with $\omega^2 = \sigma^2/\lambda$ (Theorem 27.2.2(e)). A fixed λ is a guess at the prior variance, and a bad guess is punished without limit, as the fixed-shrinkage curve in Figure 27.5.1 shows; choosing λ from the data is the regression version of James–Stein. With a general design the g -prior of Proposition 7.6.3(c) shrinks $\hat{\boldsymbol{\beta}}$ by the common factor $g/(1 + g)$, and estimating that factor gives the next corollary.

27.5.6 Shrinking a regression with unknown variance

In regression σ^2 is unknown but estimated independently of $\hat{\boldsymbol{\beta}}$, which is enough.

Corollary 27.5.5. James–Stein for regression

Assume the normal linear model with $\mathbf{X}$ of full column rank $p \geq 3$, let SSE have m degrees of freedom (so $\text{SSE}/\sigma^2 \sim \chi^2(m)$, independent of $\hat{\boldsymbol{\beta}}$), and for $c > 0$ let

$$\tilde{\boldsymbol{\beta}}_c = \left(1 - \frac{c \text{SSE}}{\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}}\right) \hat{\boldsymbol{\beta}}.$$

Then, with $\mathbf{Z} = (\mathbf{X}^\top \mathbf{X})^{1/2} \hat{\boldsymbol{\beta}}$,

$$E[(\tilde{\boldsymbol{\beta}}_c - \boldsymbol{\beta})^\top \mathbf{X}^\top \mathbf{X} (\tilde{\boldsymbol{\beta}}_c - \boldsymbol{\beta})] = p \sigma^2 - c m \{2(p - 2) - c(m + 2)\} \sigma^4 E\|\mathbf{Z}\|^{-2}.$$

So $\tilde{\boldsymbol{\beta}}_c$ dominates least squares in this loss iff $0 < c < 2(p - 2)/(m + 2)$, and the best constant is $c = (p - 2)/(m + 2)$. In terms of the F statistic $F = (\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}/p)/(\text{SSE}/m)$ for the hypothesis $\boldsymbol{\beta} = \mathbf{0}$, the shrinkage factor is $1 - cm/(pF)$.

Proof. By Theorem 7.5.1, $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$ and SSE is independent of it. So $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\theta} = (\mathbf{X}^\top \mathbf{X})^{1/2} \boldsymbol{\beta}$, and the loss is $\|(1 - a/\|\mathbf{Z}\|^2)\mathbf{Z} - \boldsymbol{\theta}\|^2$ with $a = c \text{SSE}$.

Conditionally on SSE, a is a constant and the computation in the proof of [Theorem 27.5.2\(a\)](#), with $c\sigma^2$ replaced by a , gives the conditional risk $p\sigma^2 - \{2a(p-2)\sigma^2 - a^2\} E\|\mathbf{Z}\|^{-2}$. Now $E(a) = cm\sigma^2$ and $E(a^2) = c^2\sigma^4 E\{\chi^2(m)^2\} = c^2m(m+2)\sigma^4$ (Theorem 4.1.2(b)). Taking expectations gives the formula, and $c\{2(p-2) - c(m+2)\}$ is positive iff $0 < c < 2(p-2)/(m+2)$ and largest at the midpoint. The last statement is $\text{SSE}/(\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}) = m/(pF)$. ■

The factor $1 - (p-2)m/\{(m+2)pF\}$ is close to $1 - 1/F$ when p and m are large: a regression that clearly fits is barely shrunk, and one indistinguishable from noise is shrunk hard. The loss concerns fitted values, not individual coefficients. Shrinking towards the fit of a reduced model, using its F statistic, works in the same way when the full model has at least three more parameters ([Exercise \(C1\)](#)).

Example. James–Stein for Longley's regression

For Longley's six slopes there are $m = 9$ residual degrees of freedom and the overall F statistic is 330.3. The James–Stein factor with $c = (p-2)/(m+2)$ is 0.99835: the regression fits far too well for shrinkage towards zero to help. The script therefore also simulates responses at Longley's design with the least squares estimates divided by 60 as the truth, a weak signal whose noncentrality $\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} / \sigma^2$ is about 0.6, with σ equal to Longley's s . Over 20000 simulated data sets the ratio of the James–Stein prediction risk to that of least squares is 0.501.

```
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
Zr = data.drop(columns="TOTEMP").to_numpy()
y = data["TOTEMP"].to_numpy()
n, k = Zr.shape
Xc = Zr - Zr.mean(axis=0)                # centred slopes; the
    intercept is ybar
yc = y - y.mean()
b = np.linalg.lstsq(Xc, yc, rcond=None)[0]
m = n - k - 1                            # residual degrees of freedom
sse = np.sum((yc - Xc @ b) ** 2)
fit_ss = b @ Xc.T @ Xc @ b               # b^T X^T X b, the regression
    sum of squares
F = (fit_ss / k) / (sse / m)
factor = 1 - (k - 2) / (m + 2) * sse / fit_ss
print(f"F = {F:.1f}; James–Stein factor {factor:.5f} = 1 - (k-2) m / ((m+2) k F)")
```

[Run this code online](#)

27.5.7 Exercises

27.5.7.1 A. Check your understanding

Exercise. A1

Show that $E\{1/\chi^2(m)\} = 1/(m-2)$ for $m \geq 3$ and that the expectation is infinite for $m \leq 2$.

Deduce that the risk of the James–Stein estimator at $\boldsymbol{\theta} = \mathbf{0}$ is $2\sigma^2$ for every $p \geq 3$, however large p is.

Solution. With the $\chi^2(m)$ density $x^{m/2-1}e^{-x/2}/\{2^{m/2}\Gamma(m/2)\}$,

$$\mathbb{E} \frac{1}{\chi^2(m)} = \frac{\int_0^\infty x^{m/2-2}e^{-x/2} dx}{2^{m/2}\Gamma(m/2)} = \frac{2^{m/2-1}\Gamma(m/2-1)}{2^{m/2}\Gamma(m/2)} = \frac{1}{2(m/2-1)} = \frac{1}{m-2},$$

where the integral converges at zero iff $m/2 - 2 > -1$, that is, $m > 2$. At $\boldsymbol{\theta} = \mathbf{0}$, $\|\mathbf{Z}\|^2/\sigma^2 \sim \chi^2(p)$, so Eq. 27.5.2 gives $p\sigma^2 - (p-2)^2\sigma^2/(p-2) = 2\sigma^2$.

27.5.7.2 B. Practice

Exercise. B1

- (a) Show that for a fixed $\boldsymbol{\theta}_0$, the estimator $\boldsymbol{\theta}_0 + (1 - (p-2)\sigma^2/\|\mathbf{Z} - \boldsymbol{\theta}_0\|^2)(\mathbf{Z} - \boldsymbol{\theta}_0)$ has risk $p\sigma^2 - (p-2)^2\sigma^4 \mathbb{E}\|\mathbf{Z} - \boldsymbol{\theta}_0\|^{-2}$. (b) Lindley's estimator shrinks towards the grand mean: $\hat{\theta}_i = \bar{Z} + (1 - (p-3)\sigma^2/S)(Z_i - \bar{Z})$ with $S = \sum_i (Z_i - \bar{Z})^2$. Show that it dominates $\mathbf{Z}$ when $p \geq 4$.

Solution.

- (a) $\mathbf{Z} - \boldsymbol{\theta}_0 \sim N_p(\boldsymbol{\theta} - \boldsymbol{\theta}_0, \sigma^2 \mathbf{I})$, and the estimation error is that of the James–Stein estimator of $\boldsymbol{\theta} - \boldsymbol{\theta}_0$ from $\mathbf{Z} - \boldsymbol{\theta}_0$; apply Eq. 27.5.2.
- (b) Choose an orthogonal $\mathbf{Q}$ whose first row is $\mathbf{1}^\top/\sqrt{p}$. In the coordinates $\mathbf{w} = \mathbf{Q}\mathbf{Z}$, the first coordinate is $\sqrt{p}\bar{Z}$ and the remaining $p-1$ are a N_{p-1} vector with squared length S . The estimator leaves the first coordinate alone and applies James–Stein in dimension $p-1 \geq 3$ to the others, so its risk is σ^2 plus less than $(p-1)\sigma^2$ (Theorem 27.5.2).

Exercise. B2

Let p be large and $\boldsymbol{\theta} = (\sqrt{p}\sigma, 0, \dots, 0)^\top$. Using the approximation $\|\mathbf{Z}\|^2 \approx \|\boldsymbol{\theta}\|^2 + p\sigma^2$, show that the James–Stein estimate of θ_1 has mean squared error of roughly $p\sigma^2/4$, far above the σ^2 of Z_1 . Why does this not contradict Theorem 27.5.2?

Solution. With $\|\mathbf{Z}\|^2 \approx 2p\sigma^2$, the factor is about $1 - (p-2)/(2p) \approx \frac{1}{2}$, so $\hat{\theta}_1 \approx Z_1/2$, with bias $-\theta_1/2$ and squared bias $p\sigma^2/4$. Halving the other $p-1$ coordinates, whose means are zero, cuts each of their risks from σ^2 to about $\sigma^2/4$. The total is roughly $p\sigma^2/4 + (p-1)\sigma^2/4 \approx p\sigma^2/2 < p\sigma^2$. One coordinate pays heavily for the others' gains; the theorem only controls the sum.

Exercise. B3

In [Proposition 27.5.4](#), show that the ratio of the Bayes risk of James-Stein to that of the Bayes rule is $1 + 2\sigma^2/(p\omega^2)$. When is the cost of estimating ω^2 negligible?

27.5.7.3 C. Going deeper

Exercise. C1

Let $\mathbf{P}$ be the orthogonal projection onto a fixed q -dimensional subspace of $\mathbb{R}^p$, with $p - q \geq 3$. Show that $\mathbf{P}\mathbf{Z} + (1 - (p - q - 2)\sigma^2/\|(\mathbf{I} - \mathbf{P})\mathbf{Z}\|^2)(\mathbf{I} - \mathbf{P})\mathbf{Z}$ dominates $\mathbf{Z}$, and translate this into shrinking a regression fit towards a nested reduced model using its F statistic.

Exercise. C2

Let $\hat{\boldsymbol{\theta}} = \mathbf{Z} + \mathbf{g}(\mathbf{Z})$ with $\mathbf{g}$ satisfying the conditions of [Lemma 27.5.1](#). Show that

$$\hat{R} = p\sigma^2 + \|\mathbf{g}(\mathbf{Z})\|^2 + 2\sigma^2 \sum_i \frac{\partial g_i}{\partial z_i}(\mathbf{Z})$$

is unbiased for the risk of $\hat{\boldsymbol{\theta}}$ (Stein's unbiased risk estimate). Compute it for James-Stein and for soft thresholding $\hat{\theta}_i = S_\lambda(Z_i)$; how could it choose λ ?